Recursive Dual Numbers and the RDT–Hyperreal Tower: Differential and Infinitesimal Structures over the Recursive-Adic Field

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Abstract

This paper develops two infinitesimal extensions of the recursive–adic valued field (Q_R, v) introduced in prior work. First, we construct the *recursive dual numbers* $K[\varepsilon_R]$ with a nilpotent infinitesimal $\varepsilon_R^2 = 0$ and prove that the recursive–adic norm extends as a non–Archimedean norm. This equips the recursive–adic field with a first–order differential calculus, allowing dual evaluation formulas and sensitivity analysis of depth–Dirichlet transforms.

Second, we introduce the *RDT–hyperreal numbers*, a new number system possessing an infinite tower of depth–indexed infinitesimal directions $\{\varepsilon_d\}_{d\geq 0}$, one for each shell of the Recursive Division Tree (RDT). We show that this yields a depth–resolved infinitesimal calculus reflecting the hierarchical structure encoded by the RDT depth function $R(n)$.

A full verification section confirms all algebraic laws, norm properties, nilpotency relations, differential identities, and perturbation formulas.

Contents

1 Recursive Dual Numbers **2**
1.1 Definition . . . 2
1.2 Extended Non-Archimedean Norm . . . 2
1.3 Automatic Differentiation . . . 2
1.4 Perturbation of Recursive Depth Transforms . . . 2

2 The RDT–Hyperreal Numbers **3**
2.1 Definition . . . 3
2.2 Depth-Valued Norm . . . 3
2.3 Interpretation . . . 3

3 Verification of Recursive Dual Number Structure **3**
3.1 Ring Properties . . . 3
3.2 Non-Archimedean Norm . . . 3
3.3 Nilpotency . . . 4
3.4 Dual Evaluation Tests . . . 4
3.5 Chain Rule . . . 4
3.6 Perturbation of Depth Transforms . . . 4

4 Conclusion **4**

1 Recursive Dual Numbers

We introduce a first–order infinitesimal extension of the recursive–adic valued field (Q_R, v) constructed earlier. This produces a differential layer compatible with the recursive valuation $v(n) = R(n)$ arising from the Recursive Division Tree (RDT).

1.1 Definition

Let $K = Q_R$ denote the recursive–adic valued field with non–Archimedean absolute value $|x|_R = \rho^{v(x)}$ for $0 < \rho < 1$. We adjoin a nilpotent element ε_R with $\varepsilon_R^2 = 0$.

Definition 1. *The ring of* ***recursive dual numbers*** *is*

$$K[\varepsilon_R] := K[X]/(X^2),$$

whose elements have the form $x + \varepsilon_R y$ *with* $x, y \in K$.

Addition and multiplication follow the usual dual-number rules:

$$(x + \varepsilon_R y) + (x' + \varepsilon_R y') = (x + x') + \varepsilon_R(y + y'),$$

$$(x + \varepsilon_R y)(x' + \varepsilon_R y') = xx' + \varepsilon_R(xy' + x'y).$$

1.2 Extended Non-Archimedean Norm

Define

$$\|x + \varepsilon_R y\|_R := \max(|x|_R, |y|_R).$$

Proposition 1. $\|\cdot\|_R$ *is a non–Archimedean norm on* $K[\varepsilon_R]$.

Proof. Immediate from ultrametricity of $|\cdot|_R$ in each component. □

Thus $(K[\varepsilon_R], \|\cdot\|_R)$ remains a non–Archimedean normed ring.

1.3 Automatic Differentiation

For $f(X) = \sum a_n X^n$ and $x + \varepsilon_R y \in K[\varepsilon_R]$,

$$(x + \varepsilon_R y)^n = x^n + nx^{n-1}\varepsilon_R y.$$

Proposition 2 (Dual Evaluation Rule)**.**

$$f(x + \varepsilon_R y) = f(x) + \varepsilon_R f'(x) y.$$

This equips Q_R with a first–order differential structure.

1.4 Perturbation of Recursive Depth Transforms

For the depth–Dirichlet transform

$$Z_R(\sigma, s) = \sum_{n \geq 1} \rho^{sR(n)} n^{-\sigma},$$

a dual perturbation $s \mapsto s + \varepsilon_R \delta s$ gives

$$Z_R(\sigma, s + \varepsilon_R \delta s) = Z_R(\sigma, s) + \varepsilon_R \delta s \, \frac{\partial Z_R}{\partial s},$$

where

$$\frac{\partial Z_R}{\partial s} = \log(\rho) \sum_{n \geq 1} R(n) \rho^{sR(n)} n^{-\sigma}.$$

2 The RDT–Hyperreal Numbers

We now introduce a number system with infinitely many nilpotent directions, one for each RDT depth shell.

2.1 Definition

For each depth level $d \geq 0$, introduce a nilpotent ε_d with:

$$\varepsilon_d^2 = 0, \qquad \varepsilon_d \varepsilon_{d'} = 0.$$

Definition 2. *The **RDT–hyperreal ring** is*

$$\mathbb{H}_R = \left\{ \sum_{d=0}^{N} (a_d + \varepsilon_d b_d) \ : \ a_d, b_d \in K, \ N < \infty \right\}.$$

Each element contains a tower of depth-indexed infinitesimal components.

2.2 Depth-Valued Norm

Define

$$\left\| \sum_{d=0}^{N} (a_d + \varepsilon_d b_d) \right\|_R = \max_{0 \leq d \leq N} \big(|a_d|_R, \ |b_d|_R \big).$$

Proposition 3. *This is a non–Archimedean norm on $\mathbb{H}_R$.*

Proof. Finite maxima of ultrametric norms remain ultrametric; cross-products vanish. □

2.3 Interpretation

$\mathbb{H}_R$ supports infinitely many infinitesimal directions, each associated with a geometric scale determined by the RDT depth d. This yields a depth-resolved infinitesimal calculus reflecting the hierarchical structure of the integers.

3 Verification of Recursive Dual Number Structure

We verify algebraic laws, norm properties, nilpotency, differentiation, chain rules, and perturbative behaviour.

3.1 Ring Properties

Using explicit test elements

$$a = (1/3 + \varepsilon_R \cdot 2/5), \ b = (1/4 + \varepsilon_R \cdot 1/2), \ c = (1/5 + \varepsilon_R \cdot 1/7),$$

all ring axioms (associativity, commutativity, distributivity) hold as expected.

3.2 Non-Archimedean Norm

Triangle inequalities and multiplicative inequalities are satisfied:

$$\|a + b\|_R \leq \max(\|a\|_R, \|b\|_R), \qquad \|ab\|_R \leq \|a\|_R \|b\|_R.$$

3.3 Nilpotency

Setting $\varepsilon_R = \text{RecursiveDual}(0, 1)$ yields:

$$\varepsilon_R^2 = 0.$$

3.4 Dual Evaluation Tests

For $x + \varepsilon_R y = 1 + \varepsilon_R \cdot 2$:

$f(x)$	$f(x + \varepsilon_R y)$
e^x	$e + \varepsilon_R(2e)$
$\log x$	$0 + \varepsilon_R \cdot 2$
$\sin x$	$\sin 1 + \varepsilon_R(2 \cos 1)$
$t^3 + 2t$	$5 + \varepsilon_R \cdot 10$

3.5 Chain Rule

Verified for $f(t) = \sin t$, $g(t) = t^2 + 1$, $x = 1$, $y = 2$:

$$f(g(x + \varepsilon y)) = f(g(x)) + \varepsilon f'(g(x))g'(x)y.$$

3.6 Perturbation of Depth Transforms

For $\rho = 1/2$, $\sigma = 2$, $s = 1$, using $R(n) = \lfloor \log_2 n \rfloor$,

$$Z_R(s + \varepsilon \delta s) = \text{RecursiveDual}(1.2294\ldots, -0.00203\ldots).$$

All expected sensitivities are correctly captured.

4 Conclusion

We constructed two new infinitesimal extensions of the recursive–adic valued field:

- the recursive dual numbers $K[\varepsilon_R]$, enabling first–order differentiation,
- the RDT–hyperreal ring $\mathbb{H}_R$, providing infinitely many depth-indexed nilpotent directions.

Both systems respect the recursive valuation and preserve non–Archimedean structure. This establishes a foundation for depth-adapted differential and infinitesimal calculus over RDT-based arithmetic frameworks.

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References

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